

## Effect of red pepper on symptoms of irritable bowel syndrome: preliminary study.

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**BACKGROUND:** Abdominal pain, that characterizes irritable bowel syndrome (IBS) together with bloating and disordered defecation, is mainly related to a visceral hypersensitivity due to an increase of TRPV(1) nociceptive nerve fiber activity. **AIM:** As capsaicin contained in red pepper is able to desensitize the TRPV(1) fibres, we evaluated whether the red pepper oral administration can decrease the symptoms of visceral hypersensitivity in IBS patients. **METHODS:** The study was performed on 50 patients with IBS diagnosed following Rome II criteria. After a 2-week washout period, 23 patients were planned to receive 4 pills/day, for 6 weeks randomly and in a double blind manner, each containing 150 mg of red pepper powder with a coat that dissolves in the colon, and 27 patients placebo. The patients scored each day in a diary the abdominal pain and bloating intensities following the 5-point Likert scale. The weekly symptom mean scores and the final patient subjective evaluation on treatment effectiveness were statistically compared among groups and intra-groups with appropriate tests. **RESULTS:** Eight patients dropped from the study: 6 in the red pepper group for abdominal pain and 2 in the placebo group. In 8 patients, the pills were reduced to 2/day, because of the abdominal pain at the onset of treatment. The intra-group comparisons showed that in patients taking red pepper the abdominal pain and bloating mean score values of the last weeks of treatment were significantly improved with respect to pre-treatment values, unlike patients taking placebo. The final patient subjective evaluation on the treatment effectiveness showed that red pepper group scored significantly better than placebo. **CONCLUSIONS:** The results of this preliminary study indicate that the chronic administration of red pepper powder in IBS patients with enteric-coated pills was significantly more effective than placebo in decreasing the intensity of abdominal pain and bloating and was considered by the patients more effective than placebo.